

Shoe Sales Data Analysis | Exploratory Data Analysis (EDA)

Project Overview

This project focuses on performing Exploratory Data Analysis (EDA) on a shoe sales dataset using Python. The goal of the project is to analyze sales patterns, product performance, customer purchasing behavior, and revenue trends to extract meaningful business insights.

The analysis was carried out using Python libraries such as Pandas, Matplotlib, and Seaborn for data manipulation and visualization.

Objectives

- Understand the structure and quality of the dataset
- Perform data cleaning and preprocessing
- Analyze sales distribution and product trends
- Identify high-performing brands and categories
- Detect outliers and unusual sales behavior
- Explore relationships between variables such as price, units sold, and revenue
- Generate business insights from the dataset

Project Workflow

1. Data Loading

- Imported dataset using Pandas
- Checked dataset shape and structure

2. Data Cleaning

- Checked for missing values
- Removed duplicates
- Verified data types

3. Exploratory Data Analysis (EDA)

Performed multiple visual and statistical analyses including:

Univariate Analysis

- Histograms
- Count plots
- Box plots
- Distribution analysis

Bivariate Analysis

- Scatter plots
- Correlation analysis
- Groupby analysis

Outlier Detection

- Boxplot visualization
- IQR method for identifying outliers

Key Insights

- Units sold have a stronger impact on revenue compared to product pricing.
- Some brands dominate the market with significantly higher product counts.
- Sales distribution appears relatively balanced across multiple product ranges.
- Certain categories and countries contribute more heavily to overall revenue.
- Outliers indicate high-value purchase behavior in specific segments.

Visualizations Included

- Distribution of Price
- Distribution of Units Sold
- Count Plot of Brands
- Revenue Analysis Charts
- Correlation Heatmaps
- Boxplots for Outlier Detection

Conclusion

This project demonstrates how Exploratory Data Analysis can be used to transform raw sales data into actionable business insights. Through visualization and statistical analysis, the project highlights important sales trends, customer behavior, and product performance metrics that can support data-driven decision-making.